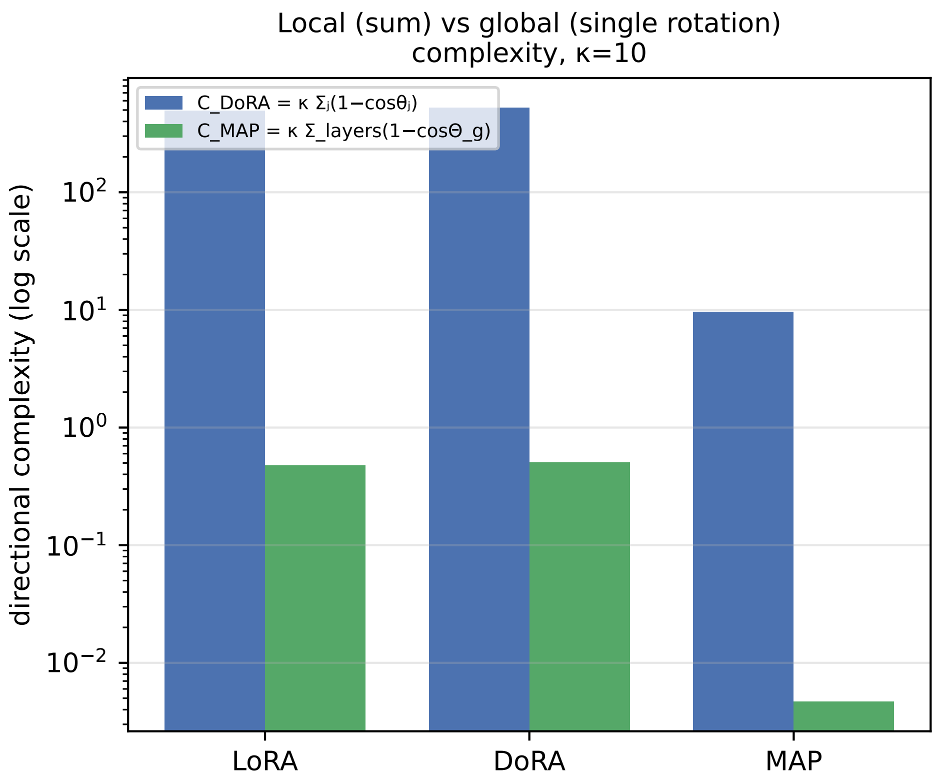
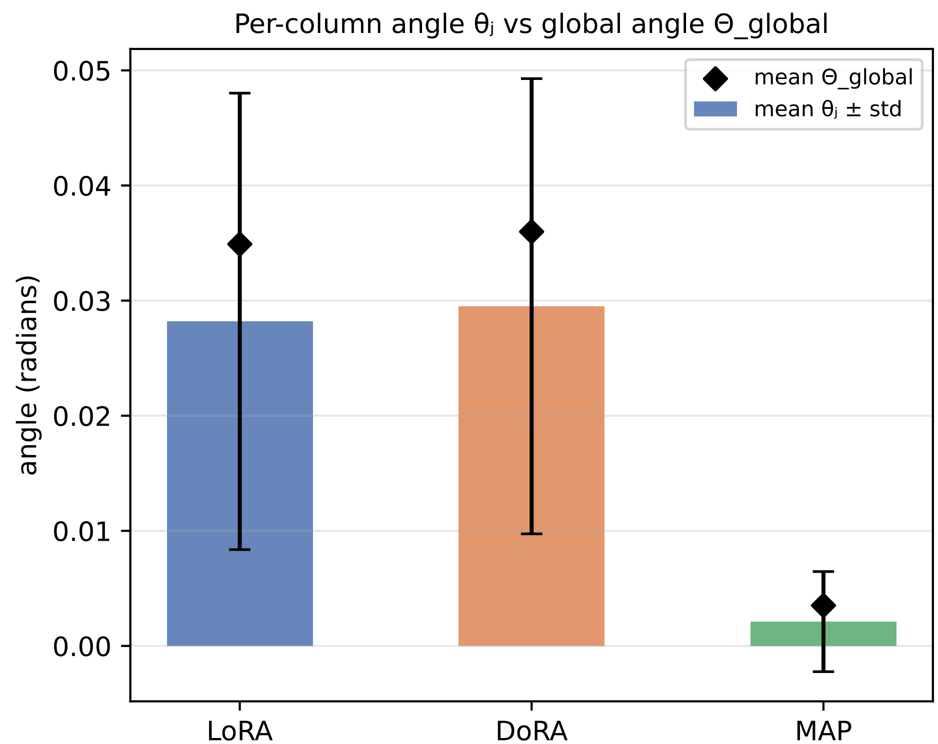
Directional Updates: LoRA vs DoRA vs MAP on RoBERTa-base / SST2

Fine-tuning metrics and aggregate directional complexity

Method	Eval acc	Trainable params	C_DoRA (Σ local)	C_MAP (global)	mean θ_j (rad)	mean Θ_g (rad)	mean s/d
LoRA	94.50%	1,919,234	492.42	0.478	0.0282	0.0349	0.931
DoRA	94.61%	2,002,178	523.04	0.507	0.0295	0.0360	0.961
MAP	92.78%	1,919,378	9.58	0.005	0.0021	0.0035	0.007





Θ_global is the single rotation of the whole flattened matrix (MAP view); θ_j are the per-column rotations (DoRA view).
Exact identity (Thm 3.10): $1-\cos\Theta_{global} = \text{mean}_j(1-\cos\theta_j)$ → DoRA sums what MAP averages.